Enhancing Traffic State Estimation at Bottlenecks through Improved Demand Modeling: A Greenshields-Grounded Approach

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Abstract

Accurate estimation of traffic state under oversaturated conditions is fundamental to a wide range of transportation applications. While intuitive, volume-to-capacity (q/q_c) ratio-based link performance functions are challenged by the U-shaped pattern of real-world speed-flow plots, which contradict the monotonic assumptions of link performance models such as the BPR function, particularly when $q/q_c \geq 1$. This study addresses this critical gap by proposing an enhanced demand estimation method grounded in the Greenshields model, incorporating a real-time inflow correction factor to more accurately capture traffic demand at bottlenecks. In parallel, a modified link performance function is introduced, replacing the traditional volume-to-capacity ratio with a demand-to-capacity (D/q_c) ratio formulation. This enables a segmented travel time and speed estimation model capable of capturing nonlinear congestion dynamics under oversaturated conditions. The effectiveness of the proposed framework is evaluated using real-world traffic data collected from two highly congested urban corridors, namely West Third Ring Road in Beijing and I-405 in Los Angeles. Results show that the proposed method significantly outperforms both the symmetric method and the quasi-density model in terms of demand and travel time estimation accuracy. In the I-405 case, for example, the mean absolute percentage error (MAPE) in travel time estimation drops from 8.54% to 3.86%, while the coefficient of determination (R^2) improves from 0.83 to 0.95. Comparative analysis further reveals that the Beijing corridor experiences higher demand, lower supply, and reduced discharge efficiency. These findings demonstrate the practical utility and robustness of the proposed method for real-world congestion diagnosis and urban mobility management.

Keywords

Traffic state estimation; Travel demand estimation; Link performance functions; Demand-to-capacity ratio; Travel time prediction; Bottleneck analysis

1 Introduction

1.1 Research background

Accurate traffic state estimation, including speed, demand, and travel time, is fundamental for transportation planning, operations, and management. Traditional estimation approaches heavily rely on link performance functions (LPFs), such as the widely used Bureau of Public Roads (BPR) function *(1)*, which estimate travel time using the volume-to-capacity (q/q_c) ratio. While effective under uncongested or moderately congested conditions, these LPFs model exhibit significant limitations when applied to oversaturated conditions where the actual demand exceeds the link's capacity.

Over the past several decades, numerous forms of LPFs have been developed *(1-6)*. Among these models, the BPR function has remained the most widely adopted due to its simplicity and ease of implementation. However, most existing LPFs, including BPR, rely heavily on the volume-to-capacity ratio as the key explanatory variable. While intuitive, this formulation raises several issues when applied to oversaturated traffic conditions, particularly in terms of how demand is defined and how the q/q_c ratio behaves when congestion causes flow to drop below demand.

In particular, bottleneck locations such as intersections, freeway ramps, and lane reductions play a central role in determining network performance *(7-8)*. Accurate quantification of travel demand at these bottlenecks is critical yet challenging due to physical constraints and latent flow behaviors. Inaccurate or biased demand estimation may introduce substantial errors in traffic state modeling, thereby limiting the effectiveness of congestion mitigation strategies and leading to suboptimal infrastructure or policy decisions. These limitations highlight the need for a more theoretically sound and practically applicable approach, particularly in high-congestion or capacity-constrained environments where accurate estimation of demand and travel time is critical.

1.2 Motivation

Building upon these challenges, the accurate estimation of bottleneck demand is particularly important for a wide range of transportation planning applications, including infrastructure design, long-term planning, and Transportation Systems Management and Operations (TSMO). In oversaturated regimes, traffic demand often becomes latent and unobservable due to physical bottlenecks, leading to a decoupling between observed volume and actual demand. This phenomenon introduces systematic bias in travel time estimation, particularly in urban networks and corridor bottleneck characterized by recurrent congestion. Existing classical static or quasi-dynamic models, which estimate demand based on observed volume, tend to underestimate true demand and fail to capture queue spillbacks, delayed vehicle entry, and flow breakdown effects.

Fig. 1 illustrates the fundamental interaction between traffic demand and supply and highlights the consequences of incorrect demand inference on traffic state estimation. In Fig. 1(a), the red demand curve shows an inverse relationship between travel cost (e.g., travel time on the Y-axis) and traveler volume (X-axis). Fig. 1(b) presents the blue supply curve, where increasing volume leads to sharply rising travel cost due to congestion and capacity constraints. Fig. 1(c) integrates these curves to identify an equilibrium point (Point B) where demand meets supply. However, when demand is misestimated, such as in oversaturated conditions where latent demand remains unobserved, the equilibrium may shift to a false point (Point A), which underrepresents both flow and travel time. This misinterpretation may result in a significant underestimation of actual

demand, thereby distorting traffic state assessment and impairing subsequent planning and management decisions.

Fig. 1 Demand interaction and equilibrium in traffic flow modeling.

A key challenge arises from the inconsistency between empirical traffic patterns and the assumptions underlying conventional LPFs. Detector-based speed-flow plots often display a U-shaped curve: speeds decrease with increasing flow until capacity is reached, followed by simultaneous declines in both speed and flow under oversaturation. However, conventional LPFs such as the BPR function assume a monotonically increasing q/q_c ratio even when $q/q_c \geq 1$, which contradicts observed behavior *(9-13)*. This mismatch raises concerns about the reliability of using q/q_c alone to model travel time under oversaturated conditions. Therefore, there is a clear need for a more robust bottleneck demand estimation approach that can accurately capture latent demand and improve traffic state estimation under oversaturated scenarios.

1.3 Objective and contributions

This study aims to enhance traffic state estimation under oversaturated conditions by developing a Greenshields-grounded demand inference method capable of capturing latent demand constrained by bottlenecks. A modified demand-delay function is also derived to reflect more accurate congestion dynamics. Together, these models support better-informed decision-making in traffic control, congestion mitigation, and urban mobility management.

This research addresses the limitations of existing demand estimation methods, particularly their inability to account for unobserved demand when the flow is constrained by bottleneck capacity. Two major challenges are targeted: (1) the inability of the volume-to-capacity ratio to represent oversaturated states accurately; and (2) the lack of a practical demand estimation framework within classical traffic flow models like Greenshields model *(14)*.

The main contributions of this study are summarized as follows:

1. Improved Demand Estimation Model. An extended Greenshields-grounded traffic flow model is proposed, incorporating a real-time inflow adjustment parameter to estimate latent demand under oversaturated conditions. This model addresses the limitations of the commonly used fundamental diagram (FD)-based symmetric method, which restricts the q/q_c ratio range and fails to account for excess demand caused by queue spillback.
2. Modified Link Performance Function. A revised demand-delay function is developed by modifying the classical BPR function to use the demand-to-capacity (D/q_c) ratio instead of the traditional q/q_c ratio. In addition, a segmented estimation model is proposed to better capture the nonlinear congestion dynamics, particularly under severe bottleneck conditions.
3. Empirical Validation and Corridor Comparison. The proposed methods are validated using real-world traffic data from two corridors: the West Third Ring in Beijing and the I-405 corridor in Los Angeles. The results confirm that the proposed model significantly improves demand and travel time estimation accuracy, with notable reductions in error metrics compared to benchmark methods. The comparative analysis also reveals differences in congestion dynamics shaped by urban structure.

The remainder of this paper is structured as follows: Section 2 reviews the related literature. Section 3 briefly presents the preliminaries that this study is built upon. Section 4 presents the

proposed methodology. Section 5 describes the case study design and data. Section 6 discusses the numerical analysis results. Section 7 concludes the paper and offers directions for future research.

2 Literature Review

In a simplified traffic system, where all demand enters upstream of a single congestion point and no other upstream bottlenecks exist, traffic demand at the bottleneck can be estimated by measuring inflow and outflow and accounting for the travel time from the observation point to the bottleneck *(15)*. However, real-world networks often contain multiple bottlenecks and exhibit congestion propagation upstream *(16)*. Under such conditions, observed flows within or downstream of the bottleneck do not accurately represent the true traffic demand, especially during oversaturated periods.

Traditionally, traffic demand estimation models are classified into static and dynamic models, based on their ability to capture temporal variation. Static models assume constant demand over time and are widely used for long-term planning. These models rely on historical data, trip generation models, and equilibrium-based traffic assignments. Examples include the Four-Step Travel Demand Model (TDM) *(17-18)* and volume-delay functions such as the BPR function *(1)*.

The BPR function and its variants form the foundation of many widely used link performance functions. The original formulation was motivated by the practical observability of traffic volume and its intuitive relationship with delay. In particular, the use of the volume-to-capacity ratio as the explanatory variable assumes a monotonic increase in travel time as volume increases, under the assumption that observed volume reflects system demand. This approach has been widely adopted in practice and research due to the ready availability of volume data from loop detectors, sensors, and traffic counts, especially when direct demand data is unavailable. It also facilitates macroscopic planning and performance assessment in large-scale networks. However, in oversaturated conditions, this assumption breaks down, as traffic volume becomes constrained by capacity and thus no longer represents true demand. This limitation motivates the need to revisit the foundational assumptions of BPR-based models and develop modified formulations that better reflect traffic dynamics under congestion. Several recent studies *(19-21)* have also emphasized this need.

While effective for macro-level planning, static models fail to capture real-time congestion dynamics such as queue spillbacks and demand surges during peak periods. In contrast, dynamic models consider time-dependent variation in demand and are commonly applied in real-time traffic control, forecasting, and adaptive management. Examples include time-dependent origin-destination (OD) estimation *(22-26)*, dynamic traffic assignment *(27)*, and machine learning-based approaches. Although dynamic models offer greater flexibility and accuracy, they require substantial real-time data and computational resources. Furthermore, both static and dynamic models often fail to capture excess demand that remains unobservable due to bottleneck capacity constraints, underscoring the need for improved demand estimation under oversaturated conditions.

One of the most common demand estimation methods relies on observed traffic flow data from loop detectors, automatic counters, or probe vehicles *(28-29)*. These methods assume that observed flow reflects actual demand. However, this assumption breaks down under oversaturated conditions, as observed flow becomes constrained by capacity and no longer scales with demand. Queuing effects further complicate matters: upstream queues can block vehicle entry into the

bottleneck segment, and the measured outflow may not reflect total incoming demand. Various correction approaches have been proposed, including backpropagation methods, microscopic simulation, and capacity-constrained models. An alternative and more robust strategy uses traffic density rather than observed flow for demand estimation *(13, 30)*. Traffic density is defined as the number of vehicles per unit distance and serves as a key variable in flow theory (e.g., the Greenshields model). Unlike flow, it remains elevated under oversaturated conditions, providing a more reliable indicator of congestion intensity and latent demand.

A number of classical and contemporary studies have advanced bottleneck demand estimation. May *(28)* introduced the "excess density" method to estimate demand at isolated bottlenecks. Building on this, Jia et al. *(31)* applied log-normal fitting to multi-day demand distributions and introduced an autoregressive error model. Huntsinger and Rouphail *(10)* estimated bottleneck demand by incorporating queuing effects at signalized intersections into the capacity constraint. Small and Verhoef *(32)* proposed a piecewise linear approach for hyper-congestion demand, framing it as "inflow demand" shaped by capacity and queuing. Newell *(33)* developed a polynomial-based model linking arrival rates and dissipation flows to estimate demand. Cheng et al. *(34)* further extended Newell's model by refining the polynomial demand formulation for oversaturated conditions.

In summary, traffic demand estimation methods vary in their assumptions, data requirements, and performance. Static models offer long-term insights but fail under dynamic congestion. Dynamic models adapt to time-varying demand but require rich data inputs. Flow-based methods are easy to implement but underestimate demand in oversaturated conditions. Density-based approaches provide better accuracy by capturing latent, unobservable demand. Table 1 summarizes the main categories and their characteristics.

Table 1. Summary of traffic demand estimation approaches.

Estimation approach	Representative studies	Key assumption	Advantages	Limitations
Static models	Small *(35)*	Demand remains constant over time	Simple, useful for long-term planning	Cannot capture dynamic congestion
Dynamic models	Newell *(33)*; Cheng et al. *(34)*; Zhou et al. *(19)*	Demand varies over time	Suitable for real-time applications	Requires extensive data collection
Flow-based methods	Moses et al. *(11)*	Observed flow represents actual demand	Easy to implement	Underestimates demand under oversaturated conditions
Density-based methods	Kucharski and Drabicki *(13)*	Traffic density better reflects congestion levels	Capture hidden demand and queueing effects	Requires indirect measurement
Queue-based method	Huntsinger and Rouphail *(10)*	Demand is the sum of demand at capacity and queue	Accounts for spillover and congestion effects	Requires accurate queue length estimation

3 Preliminaries

This section introduces the foundational concepts that this study builds upon, focusing on traffic dynamics under oversaturated conditions and the baseline demand inference method used for comparison. In particular, we present system characteristics relevant to travel delay and congestion at bottlenecks, followed by the introduction of the symmetric method for estimating traffic demand under oversaturated conditions *(11, 20, 21)*.

3.1 Characteristics of traffic flow under oversaturated conditions

From an analytical modeling perspective, traffic flow through bottlenecks exhibits distinct dynamics during congestion, including oversaturated flow, increasing delay, prolonged travel times, capacity drops, and delayed recovery. These phenomena reduce throughput at bottleneck locations and can lead to upstream congestion propagation, significantly impacting overall network performance. When traffic flow encounters a bottleneck, capacity limitations reduce vehicle speeds below the critical threshold v_c, marking the transition from free-flow state to congested conditions. If the arrival rate exceeds the discharge rate, vehicles accumulate upstream, forming an oversaturated traffic state. Below we use Fig. 2 to illustrate the traffic flow characteristics under oversaturated conditions.

Fig. 2(a) illustrates vehicle trajectories through a bottleneck segment from a space-time perspective. As vehicles traverse the bottleneck of length l, a congestion wave propagates upstream, causing them to slow to the congested speed v_μ. The yellow dotted line represents the actual trajectory, while the blue dashed line represents a hypothetical free-flow trajectory at speed v_f Under the point queue assumption, vehicles travel at v_f until they reach the bottleneck, where they experience a delay time w before continuing. Thus, total travel time is expressed below, where t_f is the free-flow travel time and w is the delay caused by congestion.

$$tt = t_f + w \tag{1}$$

The relationship between speed and flow is illustrated in Fig. 2(b) by blue and yellow curves, which represent Regions A and B, respectively. The upper part Region A represents the uncongested regime, where vehicles primarily travel at free-flow speed v_f due to relatively low traffic density. In this state, interactions between vehicles are minimal, allowing smooth movement along the roadway. As traffic density increases, the flow rate q gradually rises while vehicle speed v decreases. During this phase, the average vehicle speed remains close to v_f, and traffic operates efficiently. Although vehicle spacing decreases with increasing density, the road is not yet saturated, allowing vehicles to maintain relatively high speeds.

However, as traffic density continues to rise and approaches the critical density k_c, the traffic speed nears the critical speed v_c, and the flow rate reaches its maximum value q_c, marking the transition into the congested region (Region B of Fig. 2(b)). In this region, density surpasses the critical threshold, leading to reduced vehicle spacing, a decline in travel speed, and a fundamental shift in traffic conditions. As density further increases, the flow begins to decrease, and vehicle movement becomes more restricted, exacerbating congestion. Traffic no longer exhibits free-flow characteristics but instead transitions into states such as synchronized flow or wide-moving jams, where vehicle speeds are heavily influenced by preceding vehicles, making it difficult to maneuver freely. In the congested regime, vehicles must wait for downstream traffic to clear before proceeding, severely limiting road capacity. Consequently, the flow decreases, average speeds drop, and traffic oscillations become more pronounced, leading to congestion waves. When the

road becomes fully congested, traffic may reach a saturation state, characterized by extensive queuing and a significant loss of flow stability. Congestion tends to propagate upstream from a bottleneck, and any disturbance such as a traffic incident or a reduction in road capacity can intensify the severity and spatial extent of the congestion. In oversaturated conditions, the flow observed near bottlenecks is limited by the maximum capacity and therefore does not accurately represent the actual demand.

In uncongested conditions, referred to as Region A, the observed flow typically reflects the true demand. However, in the congested regime, referred to as Region C, the actual demand often exceeds the bottleneck capacity. This excess demand cannot be captured by observed flow due to capacity constraints. As a result, conventional traffic data from detectors significantly underestimates the true demand in Region C. To address this issue, it becomes necessary to apply alternative estimation strategies that can infer latent demand from observed traffic states. The symmetric method serves as one such approach by conceptually reflecting flow values across the capacity threshold, thereby approximating the level of unserved demand during periods of severe congestion.

Fig. 2(c) and Fig. 2(d) depict the relationships between flow and travel time, and flow and delay, respectively. In Fig. 2(c), travel time increases gradually as flow approaches capacity. Once flow exceeds q_c, travel time rises sharply due to queuing. Similarly, in Fig. 2(d) delay remains minimal below capacity but escalates once flow exceeds capacity, reflecting congestion-induced inefficiencies.

Fig. 2 Illustration of traffic flow characteristic in a bottleneck: (1) space-time vehicle trajectory in a bottleneck; (b) speed-flow and demand relationship (c) flow and travel time relationship; (d) flow and delay relationship.

In the next section, we introduce the symmetric method *(11, 20, 21)*, which attempts to infer the unobservable traffic demand during oversaturated periods (Region C of Fig. 2(b)) by estimating latent flow not captured by conventional sensors.

3.2 Deriving traffic demand using the symmetric method

3.2.1 Key parameters for demand-supply interaction

Accurately estimating traffic demand is critical for understanding bottleneck congestion, as it reflects the extent to which flow is constrained by roadway capacity. From a macroscopic perspective, demand is treated as a static quantity, whereas at the mesoscopic level, it evolves dynamically with traffic conditions. This subsection identifies and ranks key parameters influencing bottleneck demand estimation to improve the characterization of demand-supply interactions under oversaturated conditions.

Traffic demand parameters can be grouped by their direct influence on bottleneck formation, congestion propagation, and traffic performance. Saturation flow is one of the key factors in determining the operational capacity of signalized intersections, whereas indicators such as the q/q_c ratio, delay, queue length, and travel time are commonly used to evaluate situations where demand surpasses supply. Other parameters, such as arrival rate, discharge rate, and density, reflect

real-time congestion dynamics. Table 2 summarizes and prioritizes these parameters based on their relevance to bottleneck demand estimation.

Table 2. Classification of traffic demand parameters.

Priority levels	Parameters	Description
Highest	Saturation flow	Defines the upper limit of traffic flow. When demand exceeds capacity, the flow remains constant, marking the onset of congestion. The saturation flow rate represents the number of vehicles per hour per lane that could pass through a signalized intersection if a green signal was displayed for the full hour, the flow of vehicles never stopped, and there were no large headways. (HCM 2010, Chapter 4) *(42)*.
	q/q_c	A key metric for assessing congestion levels. A volume-to-capacity ratio q/q_c greater than 1 indicates oversaturation, meaning demand exceeds capacity. It cannot be directly measured in the field, nor is it a measure of traveler perceptions. Until capacity is reached (i.e., when flow breaks down on uninterrupted-flow facilities and when queues build on interrupted- or uninterrupted-flow facilities), the v/c ratio is not perceivable by travelers. (HCM 2010, Chapter 5) *(42)*.
High	Delay	Delay is an important performance measure for interrupted-flow system elements. A direct consequence of excessive demand. Delays increase significantly when demand surpasses capacity, impacting travel times and overall network performance. (HCM 2010, Chapter 4) *(42)*.
	Queue length	When demand exceeds capacity for a period of time or when an arrival headway is less than the service time (at the microscopic level) at a specific location, a queue forms. (HCM 2010, Chapter 4) *(42)*.
	Travel time	Estimating travel time becomes complex in oversaturated conditions due to stop-and-go waves and queuing effects. A demand-based travel time function is crucial for real-time analysis.
Medium	Arrival rate	Represents the actual traffic demand attempting to enter the bottleneck. Essential for understanding congestion formation.
	Queue discharge flow	Queue discharge flow represents traffic flow that has just passed through a bottleneck and, in the absence of another bottleneck downstream, is accelerating back to the facility's free-flow speed. Measures how efficiently vehicles exit the bottleneck. A critical factor in determining whether demand accumulation leads to severe congestion. (HCM 2010, Chapter 2) *(42)*.
Lower	Density	Density is the number of vehicles occupying a given length of a lane or roadway at a particular instant. A crucial macroscopic measure of congestion levels. High density correlates with reduced speeds and longer delays, affecting bottleneck dynamics. (HCM 2010, Chapter 4) *(42)*.
	Speed	Speed is defined as a rate of motion expressed as distance per unit of time, generally as miles per hour (mi/h). Speed reduction serves as a major indicator of bottleneck severity. Lower speeds correspond to increased queuing and reduced throughput. (HCM 2010, Chapter 4) *(42)*.
	Bottleneck type	Different bottleneck types (e.g., lane drops, intersections, merging areas, recurring, static, dynamic) influence how demand interacts with capacity constraints and flow restrictions.

3.2.2 Capacity adjustment via the symmetric method

Traffic demand can be defined as the sum of vehicles that successfully traverse a segment and those that are queued but intend to pass during the observation period. Accordingly, the total demand D at a bottleneck is given by:

$$D = \begin{cases} q, & when\ there\ is\ no\ congestion \\ q_c + Q_d, & where\ there\ is\ congestion \end{cases} \quad (2)$$

where q is observed traffic volum; D represents the total traffic demand of a road segment within a specific time period, measured in vehicles per hour per lane (veh/hr/ln); q_c denotes the maximum traffic volume passing through an intersection or road section within the given time period, also measured in veh/hr/ln; and Q_d represents the number of queued vehicles, measured in veh/hr/ln.

The symmetric method *(11, 20, 21)*, used in this study as a baseline for estimating latent demand under oversaturated conditions, is grounded in the fundamental diagram of traffic flow. It assumes a mirrored relationship between speed and flow across the capacity threshold q_c, treating it as the axis of symmetry. In uncongested conditions (Region A in Fig. 3), observed flow equals demand. However, in oversaturated regimes (Region B), the observed flow is capped at q_c, failing to reflect true demand. The method estimates this latent demand by reflecting the uncongested portion of the demand-speed curve across the axis, yielding the expression $D = 2q_c - q$ and constructing a monotonically decreasing demand-speed curve (Region C). While this assumption simplifies estimation, it may not capture the flat-speed behavior commonly observed in real-world urban traffic. This limitation is further discussed in Section 6.

The estimated demand of the symmetric method is calculated as:

$$D = q_c + (q_c - q) = 2q_c - q \quad (3)$$

From Eqs. (2) and (3), the number of queued vehicles can be expressed as:

$$Q_d = D - q_c = q_c - q \quad (4)$$

Eq. (3) leads to the following theoretical range of the D/q_c ratio under the symmetric method:

$$1 < \frac{D}{q_c} < 2 \quad (5)$$

However, under extreme or non-recurrent congestion scenarios such as peak hour, special events, or adverse weather conditions, actual demand may cause D/q_c ratio to exceed 2. However, the symmetric method defined in Eq. (3) fails to capture the excess demand, as it assumes a mirrored relationship between observed flow and laten traffic demand. Furthermore, its applicability is constrained by Eq. (5), which imposes an upper bound of 2 on the D/q_c ratio, limiting the method's accuracy under severe oversaturation. In Section 4, we propose an improved Greenshields-grounded method to address this limitation and enable the estimation of highly congested conditions where the D/q_c ratio exceeds 2.

Fig. 3 Mapping the demand with the observed flow based on the symmetric method.

This method is derived directly from the FD and provides a smooth transition between free-flow and oversaturated conditions. Its simplicity makes it suitable for real-time and short-term demand

estimation at bottlenecks. However, it does not account for long-term queue accumulation and assumes that queuing dissipates rapidly.

3.2.3 Time-accumulated method for traffic demand estimation

From the observational perspective, traffic demand D is defined as the total cumulative flow observed between the start of congestion time t_0 to the end of congestion time t_n in the assignment period. Mathematically, it is expressed as:

$$D = \sum_{t_0}^{t_n} q(t),\ t\epsilon[t_0, t_n] \tag{6}$$

where t_0 and t_n denote the start and end of the congestion period, and $q(t)$ is the observed flow at time t.

Fig. 4 Illustration of the time-accumulated method for demand estimation during congestion conditions.

Fig. 4 illustrates this method. During congestion, flow decreases and later recovers as traffic clears. This method accumulates flow over time to capture both instantaneous and latent demand, including vehicles delayed but still intending to pass through the bottleneck.

The time-accumulated method is effective for evaluating peak-hour conditions and long-term congestion patterns. Unlike point-in-time approaches, it reflects the total demand over the entire congested period. This provides a more comprehensive view of system performance and is particularly useful for predicting queue length, as the gap between cumulative demand and throughput indicates potential queue growth.

However, this method is more computationally intensive, requiring detailed time-series data across extended periods. Its accuracy also depends heavily on identifying congestion time boundaries $[t_0, t_n]$, which can be difficult in fluctuating traffic conditions. Moreover, it is not well suited for real-time applications, making it more appropriate for historical analysis and long-term planning.

4 Methodology

The symmetric method introduced in Eq. (3) serves as a useful baseline for estimating traffic demand under oversaturated conditions. However, it tends to systematically underestimate actual demand due to several inherent limitations. First, bottleneck flow constraints restrict the observed flow q to a maximum value of q_c, even when true demand exceeds this threshold, resulting in unobserved excess demand. Second, queue spillback effects are not captured by the observed flow, as vehicles queuing upstream of the bottleneck are not directly reflected in the data, despite being part of the true demand. Third, this method does not incorporate dynamic traffic density, which significantly affects how congestion builds and dissipates, further contributing to estimation inaccuracies.

This section introduces a methodological enhancement to improve the estimation accuracy of traffic demand under oversaturated conditions using the symmetric method.

4.1 Extending the Greenshields model for improved traffic demand estimation

We introduce a correction factor to capture unobserved latent demand. The correction is derived by leveraging the Greenshields model *(14)*, which offers a theoretically grounded and empirically validated framework for modeling traffic flow at the macroscopic level.

The corrected demand estimation model proposed in this paper is formulated as follows:

$$D = 2q_c - q + \lambda(kv_f) \tag{7}$$

where $\lambda(\cdot)$ is an inflow correction coefficient that adjusts for congestion intensity and unmeasured demand. This term is derived from the slope of the flow-density curve in the Greenshields model. This ensures that D can be accurately estimated under different congestion conditions without underestimation.

In the Greenshields model, the relationship between flow q, density k, and speed v, is illustrated as shown in the

Fig. **5**, where v_f represents the free-flow speed, v_c represents the critical speed, k_c represents the critical density, k_j represents the jam density, q_c represents the road capacity.

Fig. 5 The fundamental diagram in Greenshields model: (a) flow-density relationship; (b) speed-flow relationship; (c) speed-density relationship.

The mathematic relationships between speed-density and flow-density of the Greenshields model can be utilized as a basis for consistency correction. The equations are given as follows:

$$v = v_f \cdot (1 - \frac{k}{k_j}) \tag{8}$$

$$q = k \cdot v_f \cdot (1 - \frac{k}{k_j}) \tag{9}$$

According to the Greenshields model, under oversaturated conditions ($q > q_c$), the rate of change of flow for density can be calculated as follows:

$$\frac{dq}{dk} = v_f \cdot \left(1 - \frac{2k}{k_j}\right) \tag{10}$$

Solving for k:

$$k_c = \frac{k_j}{2} \tag{11}$$

Thus, the maximum flow occurs at:

$$q_c = v_f \cdot k_c \cdot \left(1 - \frac{k_c}{k_j}\right) = \frac{1}{4} \cdot v_f \cdot k_j \tag{12}$$

Eq. (10) represents the rate of change of flow with density, which in turn affects the unobserved demand component. When density increases beyond k_c, flow decreases, indicating that additional

latent demand is forming upstream. To correct for this underestimation, we define λ as proportional to the absolute value of the flow-density derivative:

$$\lambda = v_f \cdot \left(\frac{2k}{k_j} - 1\right) \cdot k \tag{13}$$

A larger λ indicates higher traffic flow and density, a faster drop in flow, and a greater correction in demand estimation, making it more applicable to severe oversaturated conditions.

The final corrected demand estimation formula is given as follows:

$$D = 2q_c - q + v_f \cdot \left(\frac{2k}{k_j} - 1\right) \cdot k \tag{14}$$

The behavior of correction factor λ can be categorized as follows:

(1) When traffic density is low ($k \approx 0$), $\frac{2k}{k_j} - 1 \approx -1$, and since k is small, the correction term is close to zero. In this case, the estimated demand remains largely based on observed flow, meaning no additional correction is needed.

(2) When traffic density approaches the critical density ($k \approx \frac{k_j}{2}$), $\frac{2k}{k_j} - 1 \approx 0$, making the correction term zero. This means that the Greenshields model accurately reflects flow conditions, and no extra correction is required.

(3) When traffic density is high ($k \to k_j$), $\frac{2k}{k_j} - 1$ becomes positive and large, and since k is also large, the correction term increases significantly. This accounts for the fact that observed flow underestimates actual demand because vehicles are restricted by congestion and cannot pass through the bottleneck freely.

Specifically, to accurately estimate traffic demand at bottleneck locations, we propose the following solution process.

(1) First, we calibrate Greenshields model parameters by calibrating v_f, k_j and q_c using historical data.

(2) Next, we compute the observed flow q by extracting roadway flow data from traffic sensors. Then, we determine whether the traffic condition is oversaturated: if $q \leq q_c$, travel time is directly calculated; if $q > q_c$, a corrected formula is applied to estimate demand D and incorporate it into the demand-delay function.

(3) Finally, we estimate travel time and perform adjustments by comparing the results with actual observed data.

Fig. 6 Mapping the demand with the observed flow based on the symmetric method.

The D/q_c ratio under the original symmetric method is derived according to Eq. (3):

$$\frac{D}{q_c} = \frac{2q_c - q}{q_c} = 2 - \frac{q}{q_c} \tag{15}$$

This implies that $1 < \frac{D}{q_c} < 2$, as illustrated in Fig. 6. However, in extreme or non-recurrent congestion scenarios (e.g., accidents, special events, weather disruptions), the actual demand may cause D/q_c to exceed 2.

Using the Greenshields-grounded modified demand formulation in Eq. (14), the resulting D/q_c ratio can be expressed as:

$$\frac{D}{q_c} = \frac{2q_c - q + v_f \cdot \left(\frac{2k}{k_j} - 1\right) \cdot k}{q_c} = 2 - \frac{q}{q_c} + \frac{v_f \cdot k}{q_c} \cdot \left(\frac{2k}{k_j} - 1\right), \qquad \frac{D}{q_c} > 1 \tag{16}$$

When $q = q_c$, both Eq. (14) and Eq. (15) yield $D/q_c = 1$. However, when $q > q_c$, i.e., $k > k_c$, D/q_c in Eq. (16) can exceed 2, capturing more severe oversaturation levels, as shown by the red curve in Fig. 6.

Therefore, the proposed modified approach addresses the limitations of the traditional symmetric method *(11, 20, 21)*, providing improved estimation accuracy across varying levels of resolution and enhanced applicability under both typical and extreme congestion conditions.

4.2 Derivation of travel time estimation model

In most papers related to link performance functions, researchers have primarily focused on the BPR function in its polynomial form, as it efficiently and simply relates traffic volume to delays, making it widely used in transport planning software. However, it also presents certain inherent challenges *(36-38)*. The calibrated function may exhibit a non-negligible error when applied to congestion conditions, such as bottleneck. Additionally, the BPR function fails to capture traffic flow dynamics and queue evolution processes *(39)*, which limits its accuracy and applicability in describing oversaturated bottlenecks characterized by high density but low throughput. Therefore, in this section, we use the BPR function as a benchmark and modify its key parameter, q/q_c, to enhance its accuracy for application in oversaturated conditions.

The classical BPR function is formulated as follows:

$$tt = t_f \cdot [1 + \alpha(\frac{q}{q_c})^{\beta}] \tag{17}$$

where tt is the link travel time, t_f is the travel time at the free-flow speed, q is the volume, q_c is the capacity of the link or segment, α and β are parameters, as reference value, $\alpha = 0.15, \beta = 4$.

Within traffic flow analysis, the q/q_c ratio serves as a fundamental indicator for assessing roadway capacity usage, particularly before the emergence of severe or oversaturated congestion. It is calculated as the ratio of actual traffic volume q to the maximum road capacity q_c. However, when traffic enters a state of severe congestion or oversaturation, q becomes constrained by bottlenecks, making it unable to accurately reflect the true traffic demand. In such cases, the D/q_c ratio provides a better representation of the supply-demand imbalance. D represents the number of vehicles that would ideally pass through without bottlenecks or delays and comparing it to q_c helps determine whether a road is operating beyond its capacity.

Typically, q/q_c ratio is suitable for analyzing traffic conditions before severe congestion, whereas D/q_c ratio is more appropriate for scenarios involving queue spillovers, bottleneck constraints, and

excessive demand exceeding capacity. When $D/q_c > 1$, it indicates that demand has exceeded the road maximum capacity, potentially leading to traffic delays and queue accumulation. Therefore, in severe congestion analysis and transportation planning research, D/q_c is adopted as a more accurate measure to describe the traffic demand-supply relationship.

In this study, the q/q_c ratio can be calculated using the following piecewise formula:

$$\frac{q}{q_c} = \begin{cases} \frac{q}{q_c}, q \le q_c \\ \frac{D}{q_c}, D > q_c \end{cases} \tag{18}$$

By substituting Eq. (18) into Eq. (17), we derive the modified BPR function characterizing travel time in uncongested conditions as follows:

$$tt = t_f \cdot [1 + \alpha(\frac{q}{q_c})^{\beta}] \tag{19}$$

For the oversaturated conditions, the travel time can be estimated as follows:

$$tt = t_f \cdot [1 + m(\frac{D}{q_c})^{n}] \tag{20}$$

In this study, we adopt a piecewise calibration strategy for the proposed link performance function to better reflect travel time behavior under different traffic regimes. For uncongested conditions, we retain the classical BPR functional form using the observed volume-to-capacity ratio q/q_c, with parameters α and β calibrated accordingly.

However, for oversaturated conditions, where the observed volume becomes constrained by capacity and no longer reflects actual demand, we introduce a corrected demand-to-capacity ratio D/q_c. The oversaturated regime is modeled using a modified BPR structure with parameters m and n. This formulation enables the model to capture nonlinear travel time escalation due to latent demand buildup and capacity spillback.

Since the behavioral dynamics and explanatory variables differ between the two regimes, the corresponding parameter sets, α and β for uncongested conditions, and m and n for oversaturated conditions, are calibrated separately. This segmentation is essential for improving the model's flexibility and accuracy in representing the full spectrum of traffic conditions.

Fig. 7 Monotonically increasing relationship between volume-to-capacity ratio and travel time.

Fig. 7 illustrates the curve patterns of Eqs. (18) and (19). Fig. 7(a) depicts the uncongested conditions, where travel time increases gradually as the q/q_c ratio rises. The curve remains relatively stable, exhibiting a slow growth rate over a long interval without a noticeable steep rise. In contrast, Fig. 7(b) represents the oversaturated conditions, where travel time increases significantly as the D/q_c ratio grows. The curve rises sharply, and as the D/q_c ratio increases, the growth rate accelerates further, demonstrating an exponential growth trend. Overall, the two piecewise curve exhibits a steadily increasing trend, indicating that travel time rises as the volume-to-capacity and demand-to-capacity ratio increases.

By assuming a constant length, the estimated speed can be derived from Eq. (19) and Eq. (20) as the inverse of travel time as follows:

$$v = \frac{v_f}{[1 + \alpha(\frac{q}{q_c})^{\beta}]} \tag{21}$$

Therefore, during the uncongested conditions, the estimated speed could be rewritten as follows:

$$v = v_f \cdot \frac{1}{[1 + \alpha(\frac{q}{q_c})^{\beta}]} \tag{22}$$

And under the oversaturated conditions, the speed could be estimated as follows:

$$v = v_f \cdot \frac{1}{[1 + m(\frac{D}{q_c})^{n}]} \tag{23}$$

Fig. 8 Monotonically decreasing relationship between volume-to-capacity ratio and speed.

The relationship between speed and q/q_c ratio is shown in Fig. 8. When q/q_c ratio is less than 1 (i.e., $q/q_c \leq 1$), the speed exhibits a decreasing trend, but the overall change is relatively small, as shown in Fig. 8(a). When q/q_c ratio exceeds 1 (i.e., $D/q_c > 1$), the speed undergoes significant changes, decreasing sharply as D/q_c increases, as illustrated in Fig. 8(b). Overall, the relationship between speed and q/q_c ratio tends to decrease monotonically, with $q/q_c = 1$ serving as a critical threshold.

In summary, this study derives a macroscopic traffic performance function under oversaturated conditions. When there is no traffic congestion, i.e., when $q/q_c \leq 1$, the core parameter of the travel time function is the q/q_c ratio. At this stage, the model primarily describes whether the actual traffic flow has reached the maximum road capacity. A higher q/q_c ratio indicates that the traffic flow is approaching saturation. When traffic congestion occurs, i.e., when $q/q_c > 1$, the core parameter of the function is the D/q_c ratio. At this stage, the model primarily reflects the relationship between traffic demand and road capacity. A higher D/q_c ratio indicates a greater degree of road congestion.

5 Case Study

5.1 Traffic dataset overview

To demonstrate the effectiveness and applicability of the proposed demand estimation method under oversaturated conditions, we present two representative case studies.

Case 1: The Beijing West Third Ring corridor. This corridor extends from Wanliu Bridge in the south to Suzhou Bridge in the north, spanning a total length of 14.8 km. It comprises seven interchange bridges and 21 entrance and exit ramps, serving as a major north-south traffic corridor connecting the western urban area of Beijing. Traffic data was collected from 15 detectors along the south-to-north direction of the West Third Ring Road. Multi-day data was collected continuously over 16 days, from Monday, June 15, 2021, to Wednesday, June 30, 2021, covering

both weekdays and holidays, with 24-hour monitoring each day. The collected data includes traffic flow and speed, recorded at a time interval of 2 minutes.

The layout of detector deployment and bottleneck locations on the Beijing West Third Ring corridor is shown in Fig. 9, and detailed detector information is provided in Table 3.

Fig. 9 Detector deployment and bottleneck location on the Beijing West Third Ring corridor.

Table 3. Detector information of Beijing West Third Ring corridor.

Detector ID	Name	Number of lanes	Link length (m)	Detector ID	Name	Number of lanes	Link length (m)
1	HI2092c	4	500	9	HI8062c	4	767
2	HI2090c	3	1,653	10	HI8026c	3	587
3	HI8030c	4	1,093	11	HI8025c	3	746
4	HI2065c	4	924	12	HI8024c	3	1,194
5	HI2063c	3	777	13	HI8023c	3	333
6	HI8064c	3	512	14	HI8073c	3	148
7	HI8063c	3	546	15	HI8072c	3	613
8	HI8027c	3	1,157				

Case 2: The Los Angeles (LA) I-405 corridor. This corridor is the most frequently used expressway leading to Los Angeles International Airport (LAX) and is one of the busiest roads in the United States, serving as a critical transportation artery for the West Coast region. The selected study section covers a south-to-north segment from absolute mileage 12.85 km to 23.90 km, including 22 detectors and 23 entrance and exit ramps. Traffic flow data was collected using magnetometer-based detectors. Multi-day data was collected continuously over 75 days, from Saturday, April 3, 2021, to Saturday, July 31, 2021, covering both weekdays and holidays, with 24-hour monitoring each day. The collected data includes traffic flow, speed, and occupancy, recorded at a time interval of 5 minutes.

The layout of the Los Angeles I-405 corridor and detector deployment is shown in Fig. 10, and detailed detector information is provided in Table 4.

Fig. 10 Detector deployment and bottleneck location on the Los Angeles I-405 corridor.

Table 4. Detector information of Los Angeles I-405 corridor.

Detector ID	Name	Number of lanes	Link length (mi)	Detector ID	Name	Number of lanes	Link length (mi)
1	1201735	4	0.18	12	1201541	6	0.18
2	1221594	6	0.18	13	1221434	6	0.25

3	1201705	4	0.21	14	1201510	6	0.57
4	1221482	4	0.42	15	1214461	4	0.50
5	1201671	4	0.42	16	1214080	5	0.23
6	1201637	4	0.42	17	1201481	5	0.22
7	1221470	5	0.45	18	1201453	4	0.23
8	1201606	5	0.21	19	1214238	4	0.38
9	1221451	5	0.35	20	1214212	4	0.40
10	1209243	6	0.56	21	1201430	4	0.22
11	1213650	6	0.38	22	1209204	6	0.34

Once the datasets are introduced, we proceed to calibrate fundamental diagram models using observed traffic data.

5.2 Identifying key parameters in the Greenshields model

To apply Eq. (13) for demand estimation, we calibrate key parameters of the FD model: free-flow speed v_f, critical speed v_c, critical density k_c, jam density k_j, and capacity q_c. The theoretical FD often deviates from observed data, necessitating calibration to enhance the model's applicability and accuracy, thereby improving its consistency with real-world traffic conditions. By calibrating with observed data, key parameters can be accurately estimated, thereby enhancing the reliability of traffic estimation, simulation, optimization models, and management strategies. The FD can be expressed in terms of the flow-density relationship, speed-density relationship, or flow-speed relationship. These relationships are interconnected by the equation $q = kv$, where q is the flow, v is the speed, and k is the density.

We calibrate the parameters using the Least Squares Method (LSM), minimizing the objective function:

$$J = min \sum_{i=1}^{n} [y_i - f(x_i, v_f, v_c, k_c, k_j, q_c)]^2 \quad (24)$$

where y_i are the observed flow, speed and density values, $f(x_i, v_f, v_c, k_c, k_j, q_c)$ are the model estimations, v_f, v_c, k_c, k_j, q_c are the model parameters, and n is the number of observed points.

Using this method, we calibrated the key parameters of the Greenshields model, Van Aerde model *(40)*, and S3 models *(41)*, providing a solid foundation for subsequent traffic demand and travel time estimation. Although the Van Aerde and S3 models exhibit lower fitting errors and better overall performance, they are used primarily as benchmark models. Given that this study builds upon an extension of the Greenshields model, its calibration results are adopted for all subsequent analyses.

The fitting curves for the representative cases are shown in Fig. 11 and Fig. 12. The coefficient of determination (R^2) reaches 0.83 for the Beijing West Third Ring dataset and 0.87 for the Los Angeles I-405 dataset using the Greenshields model, indicating a strong agreement with observed data. These results confirm that the calibrated parameters are sufficiently accurate for supporting the analyses that follow. In future work, we will consider refining alternative traffic flow models and improving demand estimation methods.

Table 5 summarizes the key calibrated parameters for the two cases across different traffic flow models. This section focuses solely on the Greenshields model. For the Beijing West Third Ring dataset, the free-flow speed is 62 km/h, with a critical speed of 31 km/h and a critical density of 45 veh/km/ln, resulting in a capacity of 1,380 veh/h/ln. The jam density is 100 veh/km/ln. In the Los Angeles I-405 corridor, the free-flow speed is slightly higher at 70 km/h. While the critical speed is higher at 35 km/h, the critical density remains the same at 50 veh/km/ln. The jam density is also 100 veh/km/ln, but the higher free-flow and critical speeds lead to an increased capacity of 1,750 veh/h/ln. This comparison suggests that the Los Angeles I-405 corridor exhibits more efficient traffic flow, as indicated by its higher capacity and critical speed, potentially reflecting better road performance or differing traffic conditions.

Table 5. The calibration of traffic flow key parameters.

Datasets	Models	v_f (km/h)	v_c (km/h)	k_c (veh/km/ln)	k_j (veh/km/ln)	q_c (veh/h/ln)
Case 1	Greenshields	62	31	45	90	1,380
	Van Aerde	70	29	43	140	1,222
	S3	68	28	43	--	1194
Case 2	Greenshields	70	35	50	100	1,750
	Van Aerde	75	40	41	115	1,645
	S3	66	37	47	--	1,710

Fig. 11 Fitting the fundamental diagram for the Beijing West Third Ring corridor: (a) flow-density relationship; (b) speed-flow relationship; (c) speed-density relationship.

Fig. 12 Fitting the Greenshields fundamental diagram for the Los Angeles I-405 corridor: (a) flow-density relationship; (b) speed-flow relationship; (c) speed-density relationship.

6 Results and Discussion

6.1 Traffic demand estimation

6.1.1 Aggregate traffic demand using multi-day data

Based on the model calibration of traffic bottlenecks in Section 4, key parameters at the bottleneck can be obtained, as shown in Table 5. Furthermore, traffic flow can be classified into congested and non-congested states using the critical speed v_c. The start and end times of congestion, t_0 and t_n, are determined by segmenting the observed time-series speed data based on the critical speed, thereby identifying key time points of the congestion period. The estimation method for traffic demand is detailed in Section 4.

Here, we introduce the average queue discharge flow during the congestion period $[t_0, t_n]$, defined as the average flow rate observed over this time interval. It is calculated as follows:

$$\mu = \frac{\sum_{t_0}^{t_n} q(t)}{t_n - t_0}, t\epsilon(t_0, t_n) \quad (25)$$

The average queue discharge flow serves as a critical indicator of bottleneck outflow capacity during congestion and provides a necessary basis for inferring latent demand and accurately characterizing oversaturated traffic conditions.

The statistical analysis of multi-day data reveals the distribution of traffic demand, D/q_c ratio, and discharge rate in the Beijing case, as shown in Fig. 13, while the results for the Los Angeles case are presented in Fig. 14. From the multi-day data distribution analysis, it is observed that in the Beijing case, the most probable traffic demand falls within the range of [3,300-3,600] veh/h/ln, accounting for 31%. The most probable D/q_c ratio corresponds to the range of [2.3-2.5], covering 38%, and 50% of the discharge rate data is distributed within [1,120-1,150]. In the Los Angeles case, the most probable traffic demand is in the range of [2,900-3,100] veh/h/ln, accounting for 16%. The most probable D/q_c ratio corresponds to the range of [1.62-1.76], covering 17%, and 19% of the discharge rate data is distributed within [1,405-1,445].

Overall, the average traffic demand during congestion is higher in the Beijing case than in the Los Angeles case, and the D/q_c ratio is also greater than that of Los Angeles corridor. This conclusion helps us better understand the differences in traffic demand during congestion in different regions. It also aids traffic managers in regulating the total traffic demand during congestion and accurately perceiving congestion severity at specific traffic bottlenecks.

Fig. 13 Distribution of traffic demand, demand-to-capacity ratio and discharge rate with weekday in Beijing West Third Ring corridor.

Fig. 14 Distribution of traffic demand, demand-to-capacity ratio and discharge rate with weekday data in the Los Angeles I-405 corridor.

Table 6. The estimation of average values of key parameters.

Datasets	t_0	t_n	P (h)	q/q_c	D/q_c	D (veh/h/ln)	q_c (veh/h/ln)	μ (veh/h/ln）
Case 1	07:27	11:45	4.28	0.47	2.33	3,213	1,380	1,123
Case 2	13:45	19:12	5.44	0.60	1.63	2,854	1,750	1,448

By averaging the data across multiple days, we derive the representative values of key parameters summarized in Table 6. It can be observed that the congestion duration at traffic bottlenecks in the Beijing case is 4.28 hours, while in the Los Angeles case, the peak-hour congestion duration is 5.44 hours. This indicates that traffic bottlenecks along the I-405 corridor in Los Angeles experience longer congestion periods than those along the West Third Ring corridor in Beijing.

As discussed earlier, the q/q_c ratio is used to evaluate road or traffic system performance during non-congested periods, while the D/q_c ratio is commonly used to assess congestion severity

during peak hours (as a dimensionless parameter). A higher D/q_c ratio indicates that the traffic demand exceeds the road capacity to a greater extent, meaning the road is less capable of accommodating vehicle flows and the congestion is more severe. As the D/q_c ratio increases, travel speeds decrease and traffic efficiency declines. Therefore, in traffic planning and management, especially during peak periods, it is essential to reasonably assess and control the D/q_c ratio based on actual conditions to optimize traffic performance and improve traffic flow quality. In this study, the average D/q_c value during congestion in the Beijing case is 2.33, while in the Los Angeles case, it is 1.63. This suggests that although congestion in Los Angeles lasts longer, its severity is lower than that in Beijing.

Furthermore, a comparison of the dynamically estimated traffic demand, road capacity, and average queue dissipation rate during congestion is shown in Table 6. In the Beijing case, the traffic demand, road capacity, and queue dissipation rate during congestion are 3,213 veh/h/ln, 1,380 veh/h/ln, and 1,123 veh/h/ln, respectively. In the Los Angeles case, these values are 2,854 veh/h/ln, 1,750 veh/h/ln, and 1,448 veh/h/ln, respectively.

In both study cases, it is evident that traffic demand during congestion far exceeds road capacity, and the queue dissipation rate during congestion is lower than road capacity in both cases. Additionally, the queue dissipation rate during congestion is lower in the Beijing case than in the Los Angeles case, while the traffic demand during congestion is significantly higher in Beijing. This further confirms that congestion in the Beijing case is particularly severe.

6.1.2 Disaggregated traffic demand estimation by weekday classification

To better manage traffic demand and control traffic flow on different weekdays, we conducted an analysis based on various weekday types. The summary of the Beijing West Third Ring case is shown in Table 7, while the summary of the Los Angeles I-405 case is presented in Table 8.

The results indicate that in the Beijing West Third Ring case, the maximum D/q_c value occurs during the Monday morning peak, reaching 2.88, with a demand of 3,972 veh/h/ln. In the Los Angeles case, the longest congestion duration occurs during the Friday evening peak, with an average duration of 6.51 hours, a D/q_c value of 1.92, and a demand of 3,363 veh/h/ln.

Table 7. Traffic parameter comparison for different weekdays on the Beijing West Third Ring corridor.

Weekday	t_0	t_n	P (h)	D/q_c	D (veh/h/ln)	μ (veh/h/ln)
Monday	07:33	11:30	3.95	2.88	3,972	1,151
Tuesday	07:11	12:37	5.43	2.17	2,990	1,137
Wednesday	07:11	11:33	4.37	2.17	2,988	1,129
Thursday	07:18	11:37	4.32	2.49	3,433	1,138
Friday	07:09	11:41	4.53	2.41	3,329	1,116

Table 8. Traffic parameter comparison for different weekdays on the Los Angeles I-405 corridor.

Weekday	t_0	t_n	P (h)	D/q_c	D (veh/h/ln)	μ (veh/h/ln）
Monday	14:14	18:47	4.55	1.38	2,416	1,535

Tuesday	14:10	19:14	5.06	1.55	2,711	1,486
Wednesday	13:46	19:12	5.43	1.57	2,747	1,444
Thursday	13:53	19:12	5.65	1.73	3,034	1,408
Friday	12:44	19:14	6.51	1.92	3,363	1,366

6.2 Travel time and speed estimation

In this section, we use LSM to estimate travel time and speed, corresponding to Eq. (26) and Eq. (27), respectively. The observed travel time data is fitted using Eq. (19) for uncongested conditions and Eq. (20) for oversaturated conditions, and the observed speed data is fitted using Eq. (22) for uncongested conditions and Eq. (23) for oversaturated.

Specifically, this involves minimizing the following objective functions for uncongested and oversaturated conditions:

$$K = min \sum_{i=1}^{I} [y_i - f(x_i, \alpha, \beta)]^2 \tag{26}$$

$$L = min \sum_{i=1}^{I} [y_i - f(x_i, m, n)]^2 \tag{27}$$

where y_i are the observed travel time and speed values, $f(x_i, \alpha, \beta)$ are the model estimations for uncongested conditions, $f(x_i, m, n)$ are the model estimation for over-saturated conditions, α, β, m, n are the model parameters, and I is the number of observed points.

We used symmetric method *(11, 20, 21)* and quasi-density method *(13, 35)* as the benchmark model and compared the travel time and speed estimation between the benchmark model and our proposed model. The evaluation metrics are the mean absolute percentage error (MAPE) and R^2. The travel time fitting results in oversaturated conditions are shown in Fig. 15. In oversaturated regions, the symmetric method yields $1 \leq D/q_c \leq 2$, the quasi-density-based criterion yields $k/k_c > 0.5$, whereas the proposed method identifies congestion when the demand-to-capacity ratio $D/q_c > 1$.

Although all three methods use the same BPR functional form, their estimation accuracy varies due to differences in how demand is modeled. The symmetric method applies a mirrored demand estimate; however, it is not suitable under oversaturated conditions as its range does not exceed 2. The quasi-density method uses density-based k/k_c, and the proposed method applies a corrected demand-to-capacity ratio D/q_c designed to account for latent demand under oversaturation. This difference in explanatory variables leads to significantly improved performance, particularly in congested regimes.

Table 9 compares the travel time performance of the symmetric method, quasi-density method, and the proposed method in two cases using MAPE and R^2 as evaluation metrics. In Case 1, the symmetric method yields a MAPE of 18.51% and an R^2 of 0.21, the quasi-density method achieves a MAPE of 5.91% and an R^2 of 0.94, while the proposed method slightly improves the performance with a reduced MAPE of 5.15% and an increased R^2 of 0.95, which indicates enhanced prediction accuracy and better model fit. In Case 2, the advantage of the proposed method becomes more pronounced. Its MAPE significantly decreases to 3.86%, compared to 8.54% for the symmetric method and 6.56% for the quasi-density method. Its R^2 also improves from 0.83 to 0.95,

demonstrating superior accuracy and improved model fit in this scenario. Overall, the proposed method outperforms the benchmark models in both cases, especially in Case 2, suggesting it is more suitable for estimating and predicting traffic demand under oversaturated conditions.

Table 9. Comparison of travel time estimation errors in oversaturated conditions.

Datasets	Methods	MAPE	R^2
Case 1	Symmetric method	18.51%	0.21
	Quasi-density method	5.91%	0.94
	Proposed method	**5.15%**	**0.95**
Case 2	Symmetric method	8.54%	0.83
	Quasi-density method	6.56%	0.88
	Proposed method	**3.86%**	**0.95**

Fig. 15 Travel time fitting results among different cases: (a) symmetric method in Beijing case; (b) quasi-density method in Beijing case; (c) proposed method in Beijing case; (d) symmetric method in LA case; (e) quasi-density method in LA case; (f) proposed method in LA case.

Table 10 compares the speed estimation errors of the benchmark methods and the proposed method in two cases. In Case 1, the symmetric method yields a MAPE of 19.92% and an R^2 of 0.13, while the quasi-density method achieves a MAPE of 6.22% and an R^2 of 0.93. The proposed method further reduces the MAPE to 5.30% and improves the R^2 to 0.96, indicating better estimation accuracy and model fit. In Case 2, the proposed method shows a more substantial advantage, with the MAPE dropping from 8.28% (symmetric method) to 4.05%, and the R^2 increasing from 0.82 to 0.96. These results suggest that while the quasi-density and proposed methods perform similarly in Case 1, the proposed method demonstrates a clear improvement in Case 2, particularly under oversaturated conditions. The fitting results for speed are illustrated in Fig. 16, which align with the patterns observed in Fig. 8(b) under over-saturated conditions.

Table 10. Comparison of speed estimation errors in oversaturated conditions.

Datasets	Methods	MAPE	R^2
Case 1	Symmetric method	19.92%	0.13
	Quasi-density method	6.22%	0.93
	The proposed method	**5.30%**	**0.96**
Case 2	Symmetric method	8.28%	0.82
	Quasi-density method	7.30%	0.86
	The proposed method	**4.05%**	**0.96**

Fig. 16 Speed fitting results among different cases: (a) symmetric method in Beijing case; (b) quasi-density method in Beijing case; (c) proposed method in Beijing case; (d) symmetric method in LA case; (e) quasi-density method in LA case; (f) proposed method in LA case.

As shown in Tables 9 and 10 and Figs. 15 and 16, the symmetric method fails to provide satisfactory estimation performance for both travel time and speed in the Beijing case. This is primarily due to its inherent limitation in representing demand when the D/q_c ratio exceeds 2, making it unsuitable for extremely congested conditions. In contrast, the proposed method, which incorporates a corrected demand-to-capacity ratio, demonstrates significantly improved accuracy and model fit. These findings further confirm the effectiveness and applicability of the proposed approach in handling oversaturated traffic scenarios.

6.3 Travel time index analysis

The Travel Time Index (TTI) is an indicator used to measure the level of traffic congestion. It represents the ratio of travel time under congested conditions to travel time under free-flow conditions. TTI is commonly used to quantify the efficiency of road systems and helps transportation planners and researchers assess the severity of traffic congestion. It can be calculated as follows:

$$TTI = \frac{tt}{t_f} \tag{28}$$

Further, we could derive the following Eq. (29) for estimate TTI for uncongested conditions and Eq. (30) for estimate TTI for oversaturated conditions.

$$TTI = 1 + \alpha(\frac{q}{q_c})^{\beta} \tag{29}$$

$$TTI = 1 + m(\frac{D}{q_c})^{n} \tag{30}$$

Table 11. The results of calibrated parameters and estimated TTI.

Datasets	Uncongested conditions			Oversaturated conditions		
	α	β	TTI	m	n	TTI
Case 1	1.30	2.05	1.28	1.33	0.67	3.34
Case 2	0.16	4.17	1.02	1.09	1.14	2.90

In Section 5.2, by fitting the proposed model with observation data, we could derive the calibrated parameters α, β, m, n, as shown in Table 11. Combined the calibrated parameters q/q_c and D/q_c in Table 6, we could estimate the TTI for the different conditions with Eq. (29) and Eq. (30). Then we could calculate the TTI in the uncongested condition is 1.28, while 3.34 for the oversaturated condition in the Beijing case. The TTI in the uncongested condition is 1.02, while 2.90 for the oversaturated condition in the LA case. Table 11 highlights the strong nonlinear impact of congestion on travel time. From the TTI values, Case 1 has a travel time index of 1.28, slightly

higher than Case 2's 1.02, indicating that Case 1 experiences more significant delays under mild congestion conditions. In the oversaturated state, congestion in Case 1 is more severe, with a travel time that is 3.34 times the free-flow state, compared to 2.90 times in Case 2. The table suggests that Case 1 is more affected by increases in traffic demand, while Case 2 exhibits a more gradual increase in travel time, showing better adaptability before severe congestion occurs.

Furthermore, from the calibrated parameters, Case 1 has $\alpha = 1.30$ and $\beta = 2.05$, indicating that its travel time is more sensitive to increases in traffic flow. In contrast, Case 2 has a higher $\beta = 4.17$, suggesting a steeper change in travel time as traffic flow increases, but with a smaller overall growth magnitude. Therefore, under high traffic flow conditions, congestion is more severe in Case 1, leading to greater travel time losses. Overall, Case 1 is more congested than Case 2, particularly in the oversaturated state, which is consistent with the conclusion in Section 5.1.

This finding demonstrates that the proposed model more accurately reflects traffic conditions when congestion occurs and better captures the relationships between traffic variables in oversaturated traffic systems. The travel time index obtained using this method is more reliable and provides a more accurate estimate of traffic congestion.

6.4 Analysis of congestion causes and regional differences

The analysis of the causes of congestion on urban expressways can be summarized from two perspectives: travel demand and road supply.

From the perspective of travel demand, the fundamental reason for congestion is that traffic demand exceeds the road supply capacity. Travel demand is reflected in the origin-destination characteristics of individual trips and the traffic volumes observed at expressway entrances and exits. Additionally, a significant number of lane-changing behaviors occur near expressway ramps and weaving areas, especially during congestion periods. These lane changes increase traffic demand, as drivers attempt to avoid long queues by switching lanes. From the perspective of road supply, urban traffic planning and the structural design of expressways also contribute to congestion. Expressway congestion is associated with several factors, including insufficient road infrastructure, structural design issues, job-housing imbalance, road network density, road hierarchy, insufficient lane capacity, excessive weaving zone length, the number of entrances and exits, spacing between ramps, and the presence of dedicated bus lanes. These factors influence the capacity and smoothness of expressway traffic flow.

The analysis of urban expressway congestion can also be categorized into dynamic and static causes. Dynamic causes of congestion include the mainline traffic volume on expressways, the traffic volume at entrance and exit ramps, and the frequency of lane changes in weaving areas. Static causes of congestion include the job-housing separation ratio, road network structure such as network density, road hierarchy, and the capacity to accommodate through traffic, roadway infrastructure including the number of mainline lanes, the length of weaving zones and transition sections, the number of interchanges, and the presence of dedicated bus lanes, as well as entrance and exit ramps in terms of their number, types, and spacing.

Table 12. Comparison of the causes of traffic congestion between two cases.

Influence Factors	Los Angeles I405	Beijing West Third Ring
Road supply capacity (veh/h/ln)	1,815	1,220

Traffic demand (veh/h/ln)	2,902	2,043
Dedicated bus lane	Yes	Yes
Job-housing separation	No	Yes
Number of mainline lanes	3	3
Number of entrance and exit ramps	23	21
Total length (km)	11.05	14.80

Table 12 summarizes several factors that may influence congestion across the two case study corridors, including road supply capacity, traffic demand, dedicated bus lanes, job-housing separation, number of mainline lanes, number of entrance and exit ramp, and total corridor length. Beijing West Third Ring Road functions as a major urban arterial that links the northern and southern parts of the city. In comparison, Los Angeles I-405 freeway offers essential access to LAX and ranks among the busiest expressways on the West Coast.

Supplementing this, Table 6 shows that traffic demand on West Third Ring corridor in Beijing is significantly higher than that on the Los Angeles I-405 corridor, while its capacity q_c is noticeably lower. Although both corridors share similar physical characteristics, such as lane count, bus lanes, and ramp frequency, they diverge in one notable aspect: the degree of job-housing separation. The I-405 corridor serves relatively balanced residential and employment zones, whereas West Third Ring corridor connects major employment centers in Zhongguancun and Shangdi with large residential communities in the Fengtai District, home to over two million people. This spatial imbalance likely contributes to intensified directional demand and congestion during peak periods.

While job-housing separation may be one contributing factor to Beijing's higher congestion levels, this interpretation remains descriptive and should not be interpreted as causal. Other contextual factors, including land use distribution, traffic signal control, and vehicle ownership rates, may also play important roles. Our current analysis does not disentangle these variables; thus, future research could employ multivariate or causal modeling approaches to quantify their individual contributions.

In summary, the comparative case study highlights the value of the proposed demand estimation method in accurately capturing travel demand and congestion levels under oversaturated conditions. The findings also underscore the need to consider regional and structural factors when interpreting traffic performance and designing targeted mitigation strategies.

6.5 Limitations of the Greenshields-based framework

While the Greenshields model offers analytical tractability and has been widely applied in macroscopic traffic flow modeling, its assumption of a linear speed-density relationship does not fully capture the nonlinear characteristics of real-world traffic, particularly under oversaturated conditions. Our empirical analysis presented in Fig. 11 shows that vehicle speeds often stabilize during congestion rather than continue to decline, forming a plateau-like region that deviates from the classical Greenshields curve.

This observation suggests that the model may not adequately reflect congested flow dynamics in dense urban networks. Although the proposed correction term improves demand estimation consistency under these conditions, it does not account for all sources of estimation error, such as variations in driver behavior, unexpected incidents, or network-level interdependencies. Therefore,

the Greenshields model should be interpreted as a first-order approximation rather than a fully realistic representation of oversaturated traffic.

To improve applicability, future work may explore more flexible formulations of the fundamental diagram, such as triangular or trapezoidal models, or integrate hybrid empirical-theoretical approaches that allow better modeling of stabilized-speed behavior and flow breakdown mechanisms.

7 Conclusions

This study proposes an improved traffic demand estimation method suitable for oversaturated traffic conditions by extending the Greenshields model with the introduction of real-time inflow rates as an adjustment parameter. A modified link performance function incorporating the D/q_c ratio is developed, along with a segmented travel time and speed estimation model. The proposed methodology is validated using real-world traffic data from the Beijing West Third Ring Road and the Los Angeles I-405 corridor. Results demonstrate that the method effectively captures latent demand and improves travel time estimation, outperforming the benchmark quasi-density method. In particular, for the Los Angeles I-405 case, the proposed approach reduces the MAPE from 8.54% to 3.86% and increases the R^2 from 0.83 to 0.95. These findings confirm the robustness and applicability of the proposed method in real-world oversaturated traffic conditions.

Furthermore, a comparative analysis of the two case studies reveals that the Beijing West Third Ring Road exhibits higher traffic demand, lower road capacity, and a lower average queue discharge rate compared to the Los Angeles I-405 corridor. A macroscopic analysis indicates that job-housing separation is the primary cause of severe congestion in the Beijing West Third Ring Road corridor, whereas the Los Angeles I-405 corridor benefits from a more balanced urban structure, leading to lower congestion levels.

By optimizing existing traffic demand estimation models, this study enhances their applicability to oversaturated traffic conditions, thereby improving the accuracy of congestion forecasting and transportation planning. Future research directions include:

(1) Artificial intelligence techniques can be integrated to enhance the accuracy of real-time traffic demand estimation. This includes leveraging deep learning models such as long short-term memory (LSTM), graph neural networks (GNN), and Transformer architectures to capture complex traffic patterns, as well as applying reinforcement learning to optimize flow control strategies.

(2) Improving the continuity of segmented estimation by employing Bayesian inference or time-series smoothing algorithms to ensure consistency in flow estimation across different road segments.

(3) Expanding the study to large-scale traffic network assignment, exploring optimization approaches based on dynamic traffic assignment, constructing cross-resolution segment flow interactions, and utilizing multi-agent reinforcement learning to optimize route selection, thereby enhancing the overall network traffic distribution balance and efficiency.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: Yuyan (Annie) Pan, Xianbiao Hu; data collection: Yuyan (Annie) Pan; analysis and interpretation of results: Yuyan (Annie) Pan; draft manuscript preparation: Yuyan (Annie) Pan, Xianbiao Hu, Qing Tang, Yanyan Chen, Xuesong (Simon) Zhou. All authors reviewed the results and approved the final version of the manuscript.

Data Availability Statement

The data used to support the findings of this study are available from the corresponding author upon request.

Conflict of Interest Statement

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